

Engineering

Give this about three hours of real work. We care about how you think, not how much you can polish. A smaller thing done well beats a bigger thing done carelessly.

The context. We are building an autonomous AI agent. A person gives it a goal in plain language, and it does the work from start to finish. It plans the steps, researches, writes and runs code, uses tools, and hands back a finished result: a report, a spreadsheet, a working page, whatever the job needed. There is a consumer app and an enterprise API. Think of it as a capable teammate that does the work, not a chatbot that gives advice.

The task: build a slice of the agent's run loop

Build a minimal but real slice of the agent: a small run loop that takes a goal, carries out a few steps using a tool, and returns a result.

We want two pieces. The first is an endpoint, roughly **POST /runs**, that accepts a goal, carries out a short sequence of steps against a mocked tool (a fake web search or file writer is perfect), tracks progress, and returns the run's status and output. The second is a tiny screen where a person types a goal, watches the steps appear, and sees the result.

You choose the stack. Keep it small. Mock the model and the tools. We care about how you handle reliability, cost, and failure, not about wiring up a real model.

Three things about the real world you have to handle.

A run can loop or run away. The agent must not execute unbounded steps and burn cost forever. Put a real limit in place and decide what happens when it is hit.

A step can fail partway. A tool call can fail after earlier steps already ran and already consumed credits. Decide what happens to the run's state and to the credits already spent. There is no single right answer, but make a call and handle it.

Cost has to be exact. Every run consumes credits. Track them precisely, with no losing count and no floating point drift.

THE SCENARIO YOUR SOLUTION NEEDS TO SURVIVE

In your walkthrough, run these against your own endpoint and show what happens. First, run a goal that completes in a few steps, and show the final output and the credits used. Then submit the exact same run request again, as if the client retried because the first response never arrived. It must not run twice or charge credits twice. Finally, run a goal where a tool step is forced to fail partway. Show what state the run ends in, what happened to the credits already

spent, and how the person could retry or recover. We want a clear, correct end state in all three cases, and for you to explain the mechanism you used, for example how you bounded the loop and how you recognized the retry.

What to submit

A link to a repo, gist, or zip with the working code and a one command way to run it. Add a short decisions file of about half a page: how you bounded the loop, how you handled the mid run failure, how you counted credits, and the one trade off you are least sure about. Include a three minute screen recording that shows the three scenarios above.

Your AI note

Include a few honest lines: what you used AI for, where you accepted its output as it was, where you overrode or corrected it and why, and one thing it got wrong that you caught. We build with AI every day and expect you to use it. What we read for is whether you were the one driving.

How we will grade

We look at five things: whether you scoped this sensibly and solved the parts that matter, whether the code is clean and actually runs, whether you handled the hard parts (bounding the loop, the partial failure, and exact credit accounting, which are the heart of it), whether you can explain and defend every line rather than letting AI think for you, and whether your decisions file and walkthrough are clear. Build the thing you would be comfortable defending.

Confidential. Please do not share this task or your submission publicly.